Semantic-CC: Boosting Remote Sensing Image Change Captioning via Foundational Knowledge and Semantic Guidance

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***Abstract*— Remote sensing image change captioning (RSICC) aims to articulate the changes in objects of interest within bitemporal remote sensing images using natural language. Given the limitations of current RSICC methods in expressing general features across multitemporal and spatial scenarios, and their deficiency in providing granular, robust, and precise change descriptions, we introduce a novel change captioning (CC) method based on the foundational knowledge and semantic guidance, which we term Semantic-CC. Semantic-CC alleviates the dependency of high-generalization algorithms on extensive annotations by harnessing the latent knowledge of foundation models, and it generates more comprehensive and accurate change descriptions guided by pixel-level semantics from change detection (CD). Specifically, we propose a bitemporal SAM-based encoder for dual-image feature extraction; a multitask semantic aggregation neck for facilitating information interaction between heterogeneous tasks; a straightforward multiscale CD decoder to provide pixel-level semantic guidance; and a change caption decoder based on the large language model (LLM) to generate change description sentences. Moreover, to ensure the stability of the joint training of CD and CC, we propose a three-stage training strategy that supervises different tasks at various stages. We validate the proposed method on the LEVIR-CC and LEVIR-CD datasets. The experimental results corroborate the complementarity of CD and CC, demonstrating that Semantic-CC can generate more accurate change descriptions and achieve optimal performance across both tasks.**

***Index Terms*— Change captioning (CC), foundation model, multitask learning (MTL), remote sensing image.**

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I. Introduction

REMOTE sensing image change interpretation technology has emerged as a pivotal tool across various domains, including environmental monitoring, urban planning, and disaster management [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. Remote sensing image change captioning (RSICC) constitutes a critical aspect of this technology, focusing on describing the changes between temporally disparate remote sensing images through natural language [11], [12], [13]. The advent of specialized datasets and the refinement of vision-language models (VLMs) have catalyzed substantial advancements in deep-learning-driven RSICC methods [13], [14].

Despite significant progress in the field of RSICC, current methodologies still struggle with performance in environments with semantic-irrelevant changes such as various lighting contrast conditions [15], [16]. This limitation is largely due to an over-reliance on deep semantic information for text generation, which tends to neglect the finer details of the images [12]. Occasionally, these methods may become ensnared by semantic noise unrelated to the change understanding task, such as shifts in the angle and intensity of lighting [3]. These factors hinder the accurate discrimination of meaningful changes. To address this issue, Liu et al. [17] have introduced a novel approach that incorporates multitask learning (MTL), where the remote sensing image change detection (RSICD) task is leveraged as a supplementary component. By harnessing pixel-level information and feeding it into the RSICC model through a shared image encoder, this strategy significantly enhances the model's ability to recognize and express effective changes, thus improving the overall accuracy.

Indeed, utilizing RSICD as an auxiliary task to enhance the semantic expression of RSICC models presents a promising approach. This is primarily due to the fact that RSICD, which encompasses the identification and precise localization of pixel-level changes, provides crucial insights that are pivotal in formulating detailed and accurate descriptions of the differences between bitemporal images. Nonetheless, integrating these two tasks within a single framework poses

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significant challenges [18], [19]. First, the scarcity of annotated datasets, which incorporate both pixel-level and semantic-level annotations, frequently leads to the overfitting of deep learning models. Second, most existing models are custom-made for a specific task. When faced with multiple tasks, these models grapple with bridging the semantic chasm between high-level semantic information and detailed pixel-level specifics. These challenges complicate the integration of homogeneous features from different tasks within a unified framework [11], [20].

Recently, vision-language foundation models have exhibited remarkable generalization capabilities, demonstrating superior performance in domain-specific transfer tasks [21], [22], [23], [24], [25], [26]. Different from the traditional neural networks, which often require meticulously designed specialized architectures to enable models to capture and represent specific features [7], [10], [27], [28], by harnessing vast amounts of data for large-scale training, foundation models can learn generalized representations, thereby efficiently transferring and sharing general knowledge across various specific downstream tasks [29], [30], [31]. The introduction of foundation models' latent knowledge can ensure performance while liberating from the dependence on extensive annotations. Moreover, large-scale VLMs possess an enhanced potential to bridge the semantic-level and pixel-level information, thereby augmenting the overall comprehension and interpretation of visual inputs [32], [33].

In this article, we introduce Semantic-CC, a novel remote-sensing image change captioning (CC) method that integrates foundational knowledge and semantic guidance. By incorporating RSICD as an auxiliary task to enhance the semantic expression of RSICC models, Semantic-CC facilitates a more comprehensive and precise interpretation of bitemporal remote sensing images. In addition, this method alleviates the dependency of high-generalization algorithms on extensive annotations by leveraging the latent knowledge inherent in foundation models. Moreover, it enhances the description of changes, making them more granular, robust, and precise through pixel-level semantic guidance from change detection (CD). Specifically, Semantic-CC consists of four components: a bitemporal SAM-based encoder for dual-image feature extraction; a multitask semantic aggregation neck for facilitating information interaction across heterogeneous tasks; a CD decoder for providing pixel-level semantic guidance; and a change caption decoder for generating descriptive sentences of changes.

The primary contributions of this article can be summarized as follows.

1) We introduce Semantic-CC, a CC method that integrates foundational knowledge and semantic guidance. This method can be trained with minimal annotations to achieve high usability and provide more granular and accurate sentence descriptions.
2) Specifically, we propose a bitemporal SAM-based encoder, which is constructed on the latent knowledge of the SAM foundation model and incorporates a bitemporal change semantic filter (BCSF) to integrate dual-temporal information into the foundational encoder. We also propose a multitask semantic aggregation neck for intra- and intertask feature attention; a straightforward multiscale CD decoder to guide the finer granular semantic expression of CC; and a CC decoding head based on Vicuna, which includes a change semantic feature enhancer that generates bitemporal differential features for the expression of change semantics.
3) To ensure the smooth convergence of CD and CC and to prevent negative transfer in MTL, we design a three-stage training strategy to supervise the training of different tasks at different stages.
4) We validate our method through experiments on the LEVIR-CD and LEVIR-CC datasets. The experimental results demonstrate the complementarity of the semantics of CC and CD and suggest that Semantic-CC can generate high-precision CD masks and CC descriptions, thereby achieving superior performance in both tasks.

II. Related Works

A. Remote Sensing Image CD

Remote sensing image CD is designed to analyze pixel-level changes between temporally distinct images, which are represented by a binary change mask, where zero signifies nonchanges and one denotes changes [3], [5], [34]. Recent methods predominantly concentrate on bitemporal remote sensing image analysis utilizing deep-learning techniques [3], [35].

Most deep learning methods for CD employ architectures based on convolution or Transformer to facilitate supervised learning from annotations, with the primary objective of discerning "effective changes". These endeavors are committed to augmenting the models' feature representation and discrimination capabilities, which specifically encompass various bitemporal feature fusion strategies [5], [36], [37], [38], multiscale feature aggregation methods [34], [39], [40], [41], and diverse spatiotemporal attention modules [35], [42], [43], [44]. For instance, Li et al. [36] proposed a temporal feature interaction module to facilitate interaction between multilevel bitemporal features of convolutional neural networks (CNNs) and to obtain multilevel differential features. Peng et al. [45] introduced a dense attention method that utilizes high-level features with category information to guide the selection of low-level features.

While supervised learning can yield quite satisfactory results and has become mainstream in CD, the requirement for a large volume of labeled data has emerged as a bottleneck for its further advancement. Some researchers have started to incorporate unsupervised learning methods and the latent knowledge of foundation models [46]. For example, Saha et al. [47] proposed a deep change vector analysis method that combines change vectors with spatial context information extracted by pretrained CNN to identify the changed pixels. Ren et al. [48] proposed a CD model based on graph convolutional networks and metric learning, which employs a multiscale dynamic graph convolutional network module to capture contextual information and extract spatial–spectral features, and combines spatial–spectral feature analysis with

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metric learning to generate reliable pseudo labels for unsupervised training.

With the evolution of large foundation models such as SAM [29], some researchers have also begun to explore the downstream applications of foundation models in the field of CD in remote sensing images. Based on the SAM, Chen et al. [3] designed a time-traveling gate module to fuse the information of pretemporal and posttemporal images, thereby enhancing the model's sensitivity to effective changes while ignoring ineffective ones. The development of foundation models has brought new possibilities to the RSICD.

The evolution of RSICD technology has led to significant achievements in the field. However, the progress of RSICD algorithms continues to encounter several challenges. First, the majority of existing RSICD datasets are characterized by low quality and limited volume. Consequently, the development of high-quality, enriched RSICD datasets has become a focal point of research. Second, the extensive use of high-resolution remote sensing images has resulted in an increased diversity of land cover types and a heightened complexity of change features. This trend necessitates the development of more effective and accurate detection methods. At last, the diverse application scenarios of RSICD have imposed new demands on efficiency.

B. Remote Sensing Image Change Captioning

RSICC represents a burgeoning multimodal task that integrates remote sensing image processing with natural language generation [19]. Currently, deep learning methods are predominantly employed to navigate the complexities inherent in this task. The foundational architecture for RSICC mirrors that utilized in the image captioning task, with both leveraging an encoder–decoder framework that incorporates CNNs or Transformers [49]. However, RSICC necessitates a concentrated focus on alterations within multitemporal images, in contrast to the image captioning task which merely requires an understanding of a singular image. The encoder in CC is tasked with extracting changed semantic features from bitemporal images, while the decoder is responsible for translating the change representation into natural language, employing technologies such as recurrent neural networks (RNNs) or Transformers.

Pioneering work by Hoxha et al. [12] and Chouaf et al. [50] involved the use of pretrained CNNs as encoders and attempted to employ RNNs and support vector machines (SVMs) as decoders to generate text descriptions. To further propel research in CC, Liu et al. [11] proposed a large-scale CC dataset, LEVIR-CC, and benchmarked several methods. They also enhanced the perception of objects of varying sizes in CC through a progressive scale perception network with a change perception layer and a scale perception enhancement module [51]. Chang and Ghamisi [15] proposed an attention network that includes a hierarchical self-attention module and a residual unit for identifying attributes related to changes and constructing semantic change embeddings. These methods, based on the traditional encoding–decoding structure, represent a classic solution for RSICC tasks. Recently, with the evolution of VLMs, high-performance RSICC methods based on these foundation models have also begun to be explored. Liu et al. [14] decoupled the task of RSICC into a binary classification task to determine the presence of changes and a text generation task, using the pretrained CLIP model and a large language model (LLM) to enhance the performance of CC. However, the exploration and application of VLMs in the RSICC field are still in their infancy, and further research into RSICC methods based on large VLMs is a promising direction.

Compared to RSICD, RSICC is a relatively nascent research direction. Over recent years, there has been a significant accumulation of technology and data in the RSICC field. Coupled with the advancement of cross-modal deep learning techniques, it has propelled RSICC to the forefront of research interest. Nevertheless, current RSICC models face challenges in generating precise, robust, and comprehensive language descriptions. This is particularly evident in the description of small targets and spatial locations. Therefore, RSICC necessitates further exploration and research.

C. Multitask Learning

MTL is a strategy that seeks to harness a singular model to manage multiple tasks concurrently. This approach has the potential to augment data utilization efficiency and mitigate overfitting through the learning of shared representations [52], [53]. Furthermore, it can bolster the robustness of the model by capitalizing on the complementary information derived from multiple tasks. This multipronged approach not only streamlines the learning process but also enhances the overall performance and reliability of the model [18].

Owing to its capacity to manage multiple tasks within a unified architecture, MTL has gained considerable traction in natural language processing (NLP) and computer vision (CV). In the NLP domain, Liu et al. [54] proposed three distinct shared information mechanisms based on RNNs to model specific tasks, yielding commendable results in four text classification tasks. Similarly, Luong et al. [55] explored three MTL modes for sequence-to-sequence models. In CV, several methodologies also concurrently learn multiple visual tasks [56], [57], [58], [59]. For instance, Chen et al. [60] proposed a large-scale MTL network with Transformers to resolve various visual tasks by fine-tuning pretrained models. Mohamed and El Sallab [61] suggested joint learning of object detection and semantic segmentation tasks. These methodologies underscore the superiority of MTL paradigms.

Beyond utilizing a single model to manage multiple subtasks, the construction of auxiliary tasks within the MTL framework to enhance the performance of the primary task has also become a prevalent paradigm [62]. Hosseinzadeh and Wang [63] devised an auxiliary network to characterize the changes between two images, thereby enabling the primary model to generate more detailed and accurate captions. Xu et al. [64] proposed an additional adaptive Transformer to accomplish the task of human motion prediction. Ak et al. [65] integrated the text-based image manipulation (TIM) task with the CC task, thereby not only enhancing the training of the TIM module but also leveraging the TIM module as additional supervision for CC training.

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MTL involves sharing parameters between tasks and learning a common representation of multiple tasks to achieve joint optimization between different tasks. However, MTL also presents certain challenges, such as conflicts and imbalances between tasks. Addressing these challenges remains a focal point of MTL research.

D. Foundation Model in Remote Sensing

LLMs have significantly advanced the progression of general artificial intelligence (AGI) and have extended their influence into the realm of remote sensing image intelligence analysis. In an effort to further enhance this field, researchers have initiated the development of large-scale multimodal VLMs for remote sensing images, aiming for more comprehensive, generalized, and stable interpretive approaches [26], [66], [67], [68]. At present, foundation models in the remote sensing domain primarily bifurcate into two research trajectories: pretraining on extensive remote sensing image data, and fine-tuning existing foundation models on a smaller scale but with high-quality data.

In the context of pretraining, several notable works have been introduced. Sun et al. [69] proposed a foundational remote sensing model framework, RingMo, which employs 2 million remote sensing image data and generative self-supervised learning for pretraining. Yao et al. [70] suggested a spatiotemporal prediction foundation model predicated on spatiotemporal evolution deblurring. Wang et al. [71] trained models including CNNs and Vision Transformers (ViT) based on the MillionAID dataset and further explored the influence of different pretraining regimes on a range of downstream tasks. Li et al. [72] introduced a VLM RS-CLIP pretrained by a pseudo-labeling technique for zero-shot remote sensing scene classification. Zhang et al. [73] propose a new framework to transfer the Foundation Model to the Remote sensing domain and present an image–text paired dataset RS5M which has five million RS images with English descriptions. Liu et al. [68] proposed the first remote sensing vision-language foundation model RemoteCLIP to learn robust visual features with rich semantics and aligned text embeddings for seamless downstream application. Large-scale pretraining necessitates substantial computing resources, thus efficient fine-tuning of large models under resource-constrained conditions warrants further investigation.

Hu et al. [67] introduced a high-quality remote sensing image captioning dataset, RSICap, and devised a scheme for fine-tuning InstructBLIP [74], proposing the remote sensing generative pretraining (RSGPT) model to swiftly align the visual features of remote sensing images with the semantic features of LLMs. Inspired by prompt learning, Chen et al. [26] designed an automated instance segmentation method based on the SAM foundation model, capable of generating suitable prompts for SAM's input, thereby enabling it to yield semantically distinguishable segmentation results for remote sensing images. In TTP [3], Chen et al. further expanded their work, designing a time-traveling gate module based on the SAM to incorporate the pre- and posttemporal information of bitemporal remote sensing images, thereby enhancing the foundation model's sensitivity to effective changes in CD tasks, while disregarding irrelevant information.

Models trained on extensive data have amassed a wealth of general knowledge, facilitating the possibility of cross-domain knowledge transfer and sharing. Large-scale pretraining and efficient fine-tuning for specific downstream tasks represent the future directions of the remote sensing image processing field and hold considerable potential for processing remote sensing images with variable spatiotemporal and scale characteristics.

III. Methodology

A. Architecture

In this article, we present Semantic-CC, an innovative method for RSICC, which is informed by the semantic guidance of CD and the prior knowledge from the foundation model. Semantic-CC employs RSICD as an auxiliary task to enhance the semantic expression of RSICC, thereby facilitating a more comprehensive and precise interpretation of bitemporal remote sensing images. The overarching structure is depicted in Fig. 1 and primarily comprises four key components: a bitemporal SAM-based encoder for the image feature extraction; a multitask semantic aggregation neck for information interaction between heterogeneous tasks; a CD decoder for supplying pixel-level semantic guidance and predicting change segmentation masks; and a change caption decoder for generating change description sentences. The entire process can be described by the following formula:

$$\begin{aligned} F_1, F_2 &= \Phi_{\text{sam-enc}}(\mathcal{I}_1, \mathcal{I}_2) \\ F_{\text{cd}}, F_{\text{cc}} &= \Phi_{\text{agg-neck}}(F_1, F_2) \\ M &= \Phi_{\text{cd-decode}}(F_{\text{cd}}) \\ T &= \Phi_{\text{cc-decode}}(F_{\text{cc}}, P) \end{aligned} \tag{1}$$

where the pre- and posttemporal images ($\mathcal{I}_1 \in \mathbb{R}^{H\times W\times 3}$ and $\mathcal{I}_2 \in \mathbb{R}^{H\times W\times 3}$) are encoded by $\Phi_{\text{sam-enc}}$ to derive bitemporal features ($F_1 \in \mathbb{R}^{(H/16)\times(W/16)\times c}$ and $F_2 \in \mathbb{R}^{(H/16)\times(W/16)\times c}$). These bitemporal features are subsequently inputted into $\Phi_{\text{agg-neck}}$ to facilitate task semantic interaction, resulting in the generation of task-sensitive features, $F_{\text{cd}} \in \mathbb{R}^{(H/16)\times(W/16)\times c}$ and $F_{\text{cc}} \in \mathbb{R}^{(H/16)\times(W/16)\times c}$, to produce the change segmentation map ($M \in \mathbb{R}^{H\times W}$), and the change caption ($T \in \mathcal{C}^N$) ($\mathcal{C}$ is the captioning vocabulary, and N is the caption length), respectively. $\Phi_{\text{cd-decode}}$ and $\Phi_{\text{cc-decode}}$ are the decoders for the CD and the CC, respectively. P symbolizes the prompt instruction of CC.

B. Bi-Temporal SAM-Based Encoder

In CD and captioning within bitemporal images, the primary objective is to discern exact changes while effectively mitigating the influence of extraneous variables such as lighting conditions, atmospheric effects, and rectification discrepancies. Foundation models have demonstrated considerable efficacy in capturing salient features within intricate scenes. However, there exists a notable limitation in the existing encoders of these foundation models, which excel in extracting features from individual images but exhibit a relative

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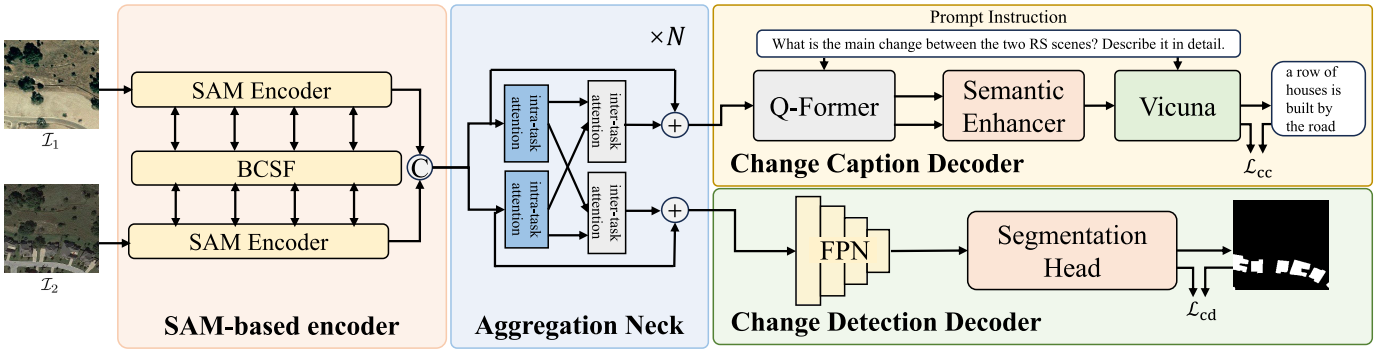


Fig. 1. Architecture of the Semantic-CC consists of four main components: a bitemporal SAM-based encoder, a multitask semantic aggregation neck, a CD decoder, and a change caption decoder.

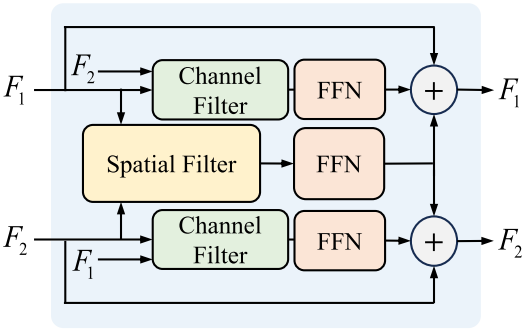


Fig. 2. Structure of the BCSF.

deficiency in processing dual images. This shortcoming is particularly pronounced when it comes to uncovering the differential characteristics inherent to bitemporal images.

To enhance the capabilities of the foundation model, thereby enabling them to discern meaningful changes between dual images and to circumvent semantic ambiguities, we have integrated a BCSF into the encoder of the SAM foundation model, as depicted in Fig. 2. The BCSF operates on both spatial and channel perspectives, thereby facilitating the transmission and filtration of change semantic features. Specifically, four instances of the BCSF have been strategically positioned after the four global attention layers within the SAM ViT encoder [75]. The functionality of the BCSF can be encapsulated by the following mathematical formulation:

$$\begin{aligned} F_1^{\text{sf}} &= \Phi_{\text{ffn}}^{\text{sf}}(\Phi_{\text{sf}}(F_1, F_2)) \\ F_2^{\text{sf}} &= \Phi_{\text{ffn}}^{\text{sf}}(\Phi_{\text{sf}}(F_2, F_1)) \\ F_1^{\text{cf}} &= \Phi_{\text{ffn}}^{\text{cf-1}}(\Phi_{\text{cf-1}}(F_1, F_2)) \\ F_2^{\text{cf}} &= \Phi_{\text{ffn}}^{\text{cf-2}}(\Phi_{\text{cf-2}}(F_2, F_1)) \\ F_k &= F_k + F_k^{\text{sf}} + F_k^{\text{cf}}, \quad k \in \{1, 2\} \end{aligned} \tag{2}$$

where Φ^{sf} and $\Phi^{\text{cf-}k}$ are used to represent the spatial and channel BCSFs, respectively. $\Phi_{\text{ffn}}^{\text{sf}}$, $\Phi_{\text{ffn}}^{\text{cf-1}}$, and $\Phi_{\text{ffn}}^{\text{cf-2}}$ refer to feed-forward networks (FFNs) that are parameterized differently. F_k^{sf} and F_k^{cf} signify the image features that result from spatial and channel filtering, respectively. $F_k \in \mathbb{R}^{(H/16)\times(W/16)\times c}$ denotes the filter's output with bitemporal interaction, with k representing the different time points.

Considering the spatial feature consistency inherent in bitemporal remote sensing images, we have adopted a shared parameter spatial filter, Φ_{sf}. However, it is recognized that different temporal phases may display unique characteristics within the channel dimension. We apply distinct channel filters for different time phases, $\Phi_{\text{cf-1}}$ and $\Phi_{\text{cf-2}}$. The spatial and channel filters are mathematically defined by the following equations:

$$\Phi_{\text{sf/cf}}(x, y) = x \otimes \sigma(\Phi_{\text{proj}}(\Phi_{\text{cat}}(x, y))) \tag{3}$$

where we leverage a straightforward methodology for the spatial/channel feature integration of the bitemporal images. In the spatial filter, x symbolizes the feature subjected to filtering, and y refers to another temporal counterpart that informs the filtering criteria. Φ_{cat} denotes feature concatenation along the channel dimension. The transformation of channel numbers from $2 \times c$ to 1 is achieved through a linear layer, Φ_{proj}. σ is the sigmoid activation function. $\otimes$ signifies the pixel-wise multiplication. The channel filter mirrors the aforementioned spatial filtering process, with the key distinction that it operates along the channel axis. The channel filter requires a preliminary step of transposing and flattening the original feature map, i.e., $H \times W \times c \to c \times (H \times W)$. The overall structure of the proposed encoder has been illustrated in Fig. 3.

It is important to highlight that, to capture the dynamics of bitemporal features efficiently, we have strategically incorporated the BCSF exclusively following the four global attention layers, leaving the other local window attention layers fixed. Moreover, given the distinct requirements for semantic level in CD versus CC tasks, we have employed two-tiered features with different channel dimensions for task decoding. Specifically, we utilize features that have undergone a dimension reduction from c to 256 for CD, while retaining the original, unreduced features for captioning.

The following mathematical formulation describes the outputs of the bitemporal SAM-based encoder:

$$\begin{aligned} F_{\text{cc}} &= \Phi_{\text{cat}}(F_1, F_2) \\ F_{\text{cd}} &= \Phi_{\text{proj}}(\Phi_{\text{cat}}(F_1, F_2)) \end{aligned} \tag{4}$$

where Φ_{proj} is dimension reduction. Φ_{cat} denotes concatenation along the batch dimension. F_{cc} and F_{cd} are features for CC and CD.

To expedite the transference of foundational knowledge acquired through extensive training on natural images, to the

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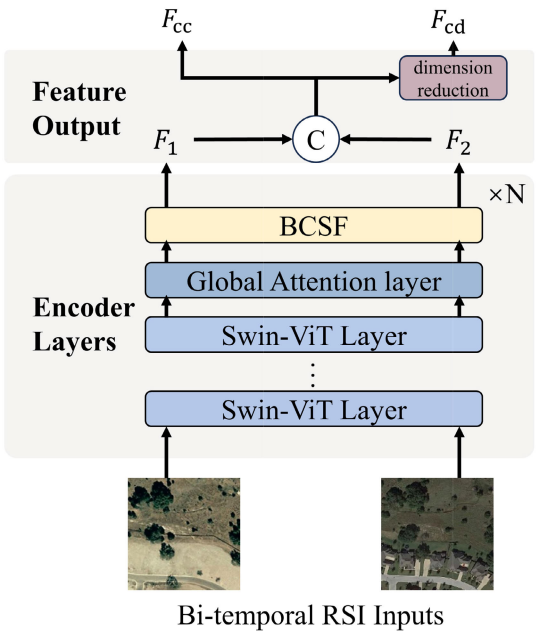


Fig. 3. Structure of Bitemporal SAM-based encoder.

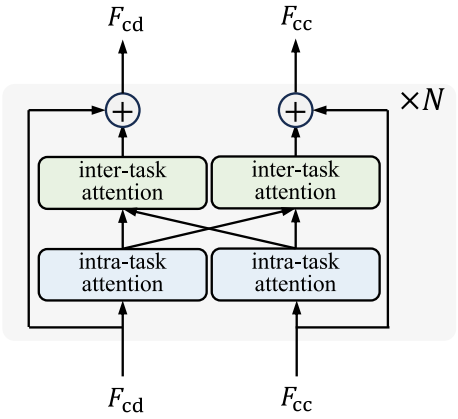


Fig. 4. Overview of the multitask semantic aggregation neck.

specialized domain of remote sensing imagery, we have integrated low-rank tunable parameters, i.e., LoRA [76], within each Transformer layer within the SAM encoder. LoRA can preserve the model's original knowledge by maintaining the majority of the existing parameters in a frozen state.

C. Multitask Semantic Aggregation Neck

The encoder of the foundation model with substantial computational load is frequently frozen during the fine-tuning for downstream tasks, impeding the convergence of MTL toward a unified latent space. Furthermore, the cross-task interaction of semantic information is crucial for MTL. In this section, we introduce the proposed multitask semantic aggregation neck to synthesize and integrate knowledge sharing across tasks. Specifically, the neck comprises N multitask semantic aggregation units, each equipped with intratask attention to mine task-specific knowledge, and intertask attention to aggregate homogeneous information from different tasks.

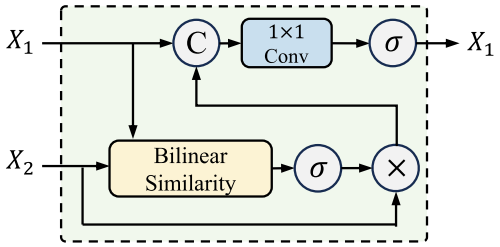


Fig. 5. Structure of the intertask attention unit.

The structure of the unit is depicted in Fig. 4 and each unit can be illustrated mathematically as follows:

$$\begin{aligned} F_{cc/cd}^{intra} &= \Phi_{att}^{intra}(F_{cc/cd}) \\ F_{cc/cd} &= F_{cc/cd} + \Phi_{att}^{inter}\left(F_{cc/cd}^{intra}, F_{cd/cc}^{intra}\right). \end{aligned} \tag{5}$$

The multitask semantic aggregation unit processes the bitemporal image features extracted by the encoder, and outputs intertask attended features. Given that the aggregation is conducted across various tasks, the task flows F_{cc} and F_{cd} from the encoder are taken as inputs. Φ_{att}^{intra} denotes intratask attention, i.e., a multihead self-attention Transformer layer. Conversely, Φ_{att}^{inter} represents intertask attention, which serves to integrate CD semantics into the CC task. Considering different channel dimensions between the features of the two tasks, we introduce an innovative cross-task attention mechanism based on bilinear similarity. The mechanism can effectively harness the homogeneous features from the heterogeneous semantic and pixel-level tasks, thereby significantly enhancing the efficacy of MTL paradigms.

The intertask attention structure, as depicted in Fig. 5, can be written by the following formula:

$$\begin{aligned} S &= X_1^T W X_2 \\ \Delta X_2 &= X_2 \otimes \sigma(S) \\ X_1 &= \sigma(\Phi_{proj}(\Phi_{cat}(X_1, \Delta X_2))) \end{aligned} \tag{6}$$

where the equation illustrates the incorporation of features from task 2 (X_2) into task 1 (X_1), i.e., $\Phi_{att}^{inter}(X_1, X_2)$. W symbolizes a matrix of learnable parameters, S indicates the computed similarity, $\otimes$ is pixel-wise multiplication, σ represents the sigmoid function, and Φ_{cat} signifies the concatenation performed along the channel dimension. Φ_{proj} projects the feature map to its original channel dimensions through a 1×1 convolution layer.

D. CD Decoder

We have proposed a streamlined multiscale CD decoder for narrowing the gap between pixel-level interpretation and extracted dense semantic features. Specifically, we concatenate bitemporal image features from the neck along the channel dimension and utilize SimpleFPN [77] to generate multiscale feature maps. These maps are then resized into a unified scale to facilitate the decoding of the change mask, as delineated in

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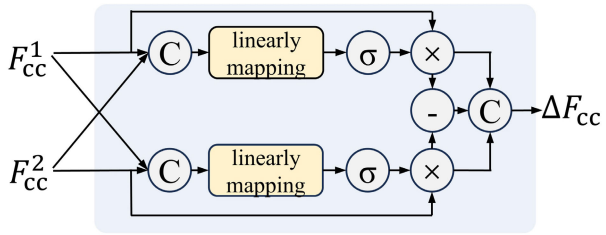


Fig. 6. Change semantic feature enhancer.

the following formulation:

$$\begin{aligned} \{F_i\} &= \Phi_{\text{sampling}}\left(\Phi_{\text{cat}}\left(F_{\text{cd}}^1, F_{\text{cd}}^2\right)\right) \\ F_i &= \Phi_{\text{resize}}(\Phi_{\text{conv}}(F_i)) \\ M &= \Phi_{\text{proj}}(\Phi_{\text{cat}}(\{F_i\})) \end{aligned} \tag{7}$$

where F_{cd}^1 and F_{cd}^2 denote the pre- and posttemporal semantic feature maps with the CD tasks, respectively. Φ_{sampling} represents the multiscale construction process based on the SimpleFPN [77]. The multiscale feature maps $\{F_i \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 512}\}$, $i \in \{1, 2, 3, 4\}$ are subsequently refined to enhance their spatial characteristics via convolution. Φ_{resize} refers to bilinear interpolation to unify the spatial resolution, while Φ_{proj} projects the channel dimension of the feature maps to the class number using a 1×1 convolution layer.

E. Change Caption Decoder

In contrast to the decoders used for vision language modeling in natural images, the CC task presents unique challenges due to the heterogeneity of the input (dual images) and the complexity of the semantics to be interpreted (effective changes). To overcome these challenges, we have developed a CC decoder based on the Vicuna LLM [78], which primarily comprises three components: a Q-Former [23] that queries semantic features associated with different temporal phases as per human instructions; a change semantic feature enhancer that generates bitemporal differential features; and a Vicuna LLM decoder that produces change descriptions. The process can be delineated by the following formula:

$$\begin{aligned} F_{\text{cc}}^{\{1,2\}} &= \Phi_{\text{q-former}}\left(F_{\text{cc}}^{\{1,2\}}, P\right) \\ \Delta F_{\text{cc}} &= \Phi_{\text{enhancer}}\left(F_{\text{cc}}^1, F_{\text{cc}}^2\right) \\ T &= \Phi_{\text{llm-decoder}}(\Phi_{\text{prompter}}(\Delta F_{\text{cc}}, P)) \end{aligned} \tag{8}$$

where the bitemporal feature maps undergo sequential processing by the Q-Former ($\Phi_{\text{q-former}}$) with shared parameters. P denotes the task-specific instructions that are meticulously designed. Φ_{enhancer} is employed to generate change features with the F_{cc}^1 and F_{cc}^2, while Φ_{prompter} serves as a template operation that combines the produced differential features into the manual instructions. $\Phi_{\text{llm-decoder}}$ procures the final change description (T). Both the Q-Former and the LLM decoder strictly conform to the structure of MiniGPT-4 [25] and inherit its pretrained weights.

The structure of the enhancer is illustrated in Fig. 6 and can be described by the subsequent formula

$$\begin{aligned} F_{\text{cc}}^{1\prime} &= F_{\text{cc}}^1 \otimes \sigma\left(\Phi_{\text{proj-1}}\left(\Phi_{\text{cat}}\left(F_{\text{cc}}^1, F_{\text{cc}}^2\right)\right)\right) \\ F_{\text{cc}}^{2\prime} &= F_{\text{cc}}^2 \otimes \sigma\left(\Phi_{\text{proj-2}}\left(\Phi_{\text{cat}}(F_{\text{cc}}^2, F_{\text{cc}}^1)\right)\right) \\ \Delta F_{\text{cc}} &= \Phi_{\text{cat}}\left(F_{\text{cc}}^{1\prime}, F_{\text{cc}}^{2\prime}, F_{\text{cc}}^{1\prime} - F_{\text{cc}}^{2\prime}\right) \\ \Delta F_{\text{cc}} &= \Phi_{\text{proj}}(\Delta F_{\text{cc}}) \end{aligned} \tag{9}$$

where Φ_{cat} symbolizes the vector concatenation along the channel dimension. $\Phi_{\text{proj-1}}$ and $\Phi_{\text{proj-2}}$ denote linearly mapping the features to a single channel number. Φ_{proj} projects the dimensions of the enhanced feature to that of the LLM's hidden state. To reconcile the semantic gap between change descriptions and natural dialog scenarios, we have integrated LoRA into the decoder of the LLM.

F. Training Strategy

Imagine we are given a training set $\mathcal{D}_{\text{train}} = \{(\mathcal{I}_1, y_1), \ldots, (\mathcal{I}_N, y_N)\}$, where $\mathcal{I}_i = \{\mathcal{I}_i^1 \in \mathbb{R}^{H\times W\times 3}, \mathcal{I}_i^2 \in \mathbb{R}^{H\times W\times 3}\}$ represents the bitemporal remote sensing images, and $y_i = \{y_i^{\text{cd}} \in \mathbb{R}^{H\times W}, y_i^{\text{cc}} \in \mathcal{C}^N\}$ denotes corresponding CD and change description annotations, respectively. However, in practical settings, we frequently encounter a shortage of annotations that concurrently encompass both y_{cd} and y_{cc}. We omit the subscript i for the sake of simplicity. This is exemplified in datasets such as LEVIR-CD [43] and LEVIR-CC [11], where merely a subset of image pairs incorporate both labels. We designate the training data that includes CD, CC, and CD & CC annotations as $\mathcal{D}_{\text{train}}^{\text{cd}}$, $\mathcal{D}_{\text{train}}^{\text{cc}}$, and $\mathcal{D}_{\text{train}}^{\text{cd-cc}}$, correspondingly. In addition, we establish the loss function for MTL as

$$\begin{aligned} \mathcal{L}_{\text{cd}} &= -\frac{1}{N}\sum_{i}^{N}(y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)) \\ \mathcal{L}_{\text{cc}} &= -\frac{1}{M}\sum_{j}^{M} w_j \log(\hat{w}_j) \\ \mathcal{L} &= \mathcal{L}_{\text{cc}} + \lambda \mathcal{L}_{\text{cd}} \end{aligned} \tag{10}$$

where $\hat{y}_i$ and y_i symbolize the prediction and annotation of an individual pixel in CD, respectively. The total number of pixels is represented by N. The one-hot encodings and the predicted caption of the jth word are denoted by w_j and $\hat{w}_j$, respectively. M is the total number of caption tokens. λ is utilized to maintain a balance in the loss.

To facilitate a smooth progression of the training process, we have formulated the subsequent training strategy: Throughout each epoch: 1) training CD with $\mathcal{D}_{\text{train}}^{\text{cd}}$; 2) training CC with $\mathcal{D}_{\text{train}}^{\text{cc}}$, keeping image encoder frozen; and 3) joint training neck with $\mathcal{D}_{\text{train}}^{\text{cd-cc}}$, keeping other parameters frozen.

IV. Experimental Results and Analyses

A. Datasets

This article investigates RSICC underpinned by the latent knowledge of the foundation model and the semantics of CD. During the experiments, the LEVIR series datasets were leveraged for both training and evaluation purposes [11], [43].

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TABLE I
COMPARISON WITH OTHER SOTA METHODS ON LEVIR-CC TEST SET

Method	METEOR	ROUGE$_L$	CIDEr	BLEU-1	BLEU-2	BLEU-3	BLEU-4
RSICCFormer [11]	0.3961	0.7412	1.3412	0.8472	0.7627	0.6887	0.6277
PromptCC [14]	0.3882	0.7372	1.3644	0.8366	0.7573	0.691	0.6354
PSNet [51]	0.3880	0.7360	1.3262	0.8386	0.7513	0.6789	0.6211
Chg2Cap [15]	0.4003	0.7512	1.3661	0.8614	0.7808	0.7066	0.6439
Semantic-CC (ours)	**0.4058**	**0.7776**	**1.3851**	**0.8807**	**0.7968**	**0.7147**	**0.6451**

These datasets encompass the CD dataset, LEVIR-CD [43], and the CC dataset, LEVIR-CC [11]. The following provides an in-depth introduction to these datasets.

1) LEVIR-CD [43]: The LEVIR-CD dataset is comprised of 637 pairs of high-resolution (VHR, 0.5 m/pixel) images obtained from Google Earth, each measuring 1024 × 1024 pixels. These bitemporal images span a period ranging from 5 to 14 years, thereby capturing significant changes in land use, particularly in relation to the construction and demolition of buildings. The fully annotated LEVIR-CD dataset encompasses a total of 31 333 instances of building changes, and we adhere to the official division of training and test sets.

2) LEVIR-CC [11]: This dataset derives its images from LEVIR-CD. The LEVIR-CC dataset modifies the bitemporal images to a size of 256 × 256 pixels, resulting in a total of 10 077 image pairs. For each image pair, five distinct annotations were collected from five different annotators to delineate the differences between the images, culminating in a total of 50 385 annotations. For our purposes, we utilized 7590 pairs for the training set, 1438 pairs for the validation set, and 2135 pairs for the test set.

B. Evaluation Metrics

For the task of CC, we employ widely used image captioning metrics such as BLEU [79], METEOR [80], ROUGE$_L$ [81], and CIDERr [82] for evaluation. The scores from these metrics are directly proportional to the similarity between the generated sentences and the reference sentences, thereby serving as an indicator of the model's performance. For the CD, our focus lies on the Precision (P), Recall (R), F1-score (F1), Intersection over Union (IoU) about the change category, and the overall segmentation accuracy (OA).

C. Implementation Details

The model presented in this study is implemented using Pytorch on the NVIDIA A100 platform. The ViT encoder of the large version SAM is utilized as the bitemporal image feature extractor, while the Vicuna-7B model serves as the large language decoder for CC. The weights for the image encoder and caption decoder are initialized from the SAM and Mini-GPT4 models, respectively. To address domain differences during training, LoRA efficient parameter fine-tuning is applied to the image encoder and caption decoder, with specific parameters including rank = 16, LoRA alpha = 32, LoRA dropout = 0.05, and no bias. The remaining parameters of the encoder, caption decoder, and Q-Former are kept frozen. The AdamW optimizer red [83] with an initial learning rate of $1e-4$ is employed for training, along with a cosine annealing scheduler with a linear warm-up to adjust the learning rate. The training process consists of 300 epochs with a batch size of 1, and the token with the highest probability is selected for output during caption generation. The number of neck units is set to 3, and the balance factor in the loss function is $\lambda = 0.5$. Various instruction prompts are used during training, such as "Describe the difference between the new remote sensing image and the old one in detail." "What is the main change between the two remote sensing scenes? Describe it in detail." "Please provide a detailed description of the difference between these two remote sensing pictures." and "Can you describe what has been changed between these two remote sensing pictures for me?" During testing, a multioutput ensemble approach is utilized to generate the final caption.

D. Comparison With the SOTA

We compare the proposed Semantic-CC with a variety of state-of-the-art (SOTA) RSICC methods, including RSICCFormer [11], PromptCC [14], PSNet [51], and Chg2Cap [15]. These methods span architectures such as CNN, Transformer, and foundation models, as well as paradigms of both single-task and MTL. The metrics for the comparative results are presented in Table I. It can be seen that Semantic-CC surpasses all SOTA methods in all metrics. The superiority is particularly noticeable in ROUGE$_L$, CIDEr, BLEU-1, BLEU-2, and BLEU-3. ROUGE$_L$, reflecting the recall, suggests that Semantic-CC can offer a more exhaustive description of scene changes, encapsulating the semantic descriptions present in the annotations. CIDEr and lower order BLEU metrics gauge the precision, indicating that Semantic-CC can accurately delineate key changes. Overall, Semantic-CC, based on foundational knowledge and semantic guidance, is capable of delivering comprehensive and precise change captions.

To provide a more intuitive representation of the effects of different methods on change sentence generation, we offer qualitative visualizations in Fig. 7. Six representative images were selected for this evaluation. The generated results reveal that as follows.

1) Owing to the pixel-level semantic guidance, the proposed method can output more precise locations and quantities. For example, in sub-figures 1 and 2, RSICCFormer and Chg2Cap inaccurately describe the

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Sample	1	2	3
Img Pair			
GT	some trees are removed and a new building appears on the bareland.	two vertical roads are constructed and two lines of completed villas appear replacing the vegetation.	a building appears next to the original one.
RSICCFormer	some houses are built in the woods.	the vegetation has been removed and a residential area with villas built along.	some houses are built on the bareland.
PromptCC	a house is built in the woods in the top left.	massive houses along the roads appear in the desert.	a building appears at the edge of the scene.
PSNet	a house is built in the woods.	the vegetation has been removed and a cross road with villas built along appears.	the scene is the same as before.
Chg2Cap	a villa appears in the woods in the middle of the scene.	the vegetation has been replaced by a road with villas built along on both sides of the road.	some houses are built on the right side of the road.
Semantic-CC (Ours)	a small house is built on the bareland and some trees are removed.	two vertical roads are built with rows of buildings along in place of the vegetation.	a completed house appears near the original house.

Sample	4	5	6
Img Pair			
GT	the scene is the same as before.	part of the woodland becomes the road and the houses on both sides of the road.	a row of detached houses are constructed beside the road on the meadow.
RSICCFormer	some houses are built along the road.	ta road with houses built along appears in the desert.	a road with houses built along appears in the desert.
PromptCC	the scene is the same as before.	many houses and a road appear in the bareland.	a road has been built and many villas have been constructed on the bareland.
PSNet	the scene is the same as before.	a road with many houses around appears in the desert.	a road with many houses on both sides appears in the desert.
Chg2Cap	some houses are built along the road.	a road with many houses on both sides of the road.	a road is built on the right side of the scene.
Semantic-CC (Ours)	the scene is the same as before.	a road with houses on both sides replaces some vegetation.	a road with detached houses is built at the bottom right corner of the area.

Fig. 7. Qualitative comparison results with other methods on the LEVIR-CC dataset, where red indicates incorrect outputs and green indicates correct outputs.

quantity and location, whereas Semantic-CC adeptly manages these complex scenarios and accurately outputs location and quantity semantics.

2) Leveraging the prior knowledge from foundation models, Semantic-CC can consistently handle complex scenarios and disturbances from illumination. For instance, in sub-figure 3, PSNet is unable to detect changes, in sub-figure 4, RSICCFormer and Chg2Cap erroneously output changes, and in sub-figure 5, other methods are unable to identify the semantic category of the forest, but Semantic-CC can still make the accurate prediction.
3) Semantic-CC can generate more standardized, fluent sentences that comply with human language systems. This is attributed to our proposed method's use of a LLM decoder, which can output change descriptions that align with human language conventions across a variety of scenarios.

E. Ablation Studies

1) Effects of Different Architecture Components: In this section, we assess the effects of the proposed components on the performance of CC. We initially evaluate the impact of the BCSF and change semantic feature enhancer (enhancer) solely on the generation of change sentences within the CC task. Subsequently, we incorporate the multitask semantic aggregation neck to broaden the scope to a MTL framework, utilizing the neck of a full Transformer architecture for comparison, i.e., intratask attention and intertask attention corresponding to the Transformer self-attention and cross-attention, respectively. The experimental results are presented in Table II.

From Table II, we can deduce the following.

1) When without semantic guidance, both the BCSF and the enhancer contribute to performance improvements, particularly on the ROUGE$_L$ metric, which gauges the model's capacity to concentrate on effective changes. The BCSF and the enhancer can extract the effective changes of bitemporal images to enhance the model's performance.
2) The model with semantic guidance outperforms that without semantic guidance across all metrics, suggesting the complementary nature of employing the pixel-level semantic guidance of the CD task to generate change captions.

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TABLE II

PERFORMANCE OF DIFFERENT ARCHITECTURE COMPONENTS ON LEVIR-CC TEST SET. BCSF: BITEMPORAL CHANGE SEMANTIC FILTER. ENHANCER: CHANGE SEMANTIC FEATURE ENHANCER. NECK_{cs} AND NECK_{sm} INDICATE TAKING THE TRANSFORMER CROSS-ATTENTION MECHANISM OR THE BILINEAR SIMILARITY MECHANISM TO FORM INTERTASK ATTENTION IN THE NECK. W/O SEMANTIC: WITHOUT SEMANTIC GUIDANCE. W/ SEMANTIC: WITH SEMANTIC GUIDANCE

Methods	BCSF	Enhancer	Neck_{cs}	Neck_{sm}	METEOR	ROUGE_L	CIDEr	BLEU-1	BLEU-2	BLEU-3	BLEU-4
w/o semantic	✗	✗	✗	✗	0.3315	0.5806	1.2254	0.76436	0.6518	0.5538	0.4657
	✓	✗	✗	✗	0.3573	0.6042	1.2992	0.8264	0.7072	0.6062	0.5275
	✓	✓	✗	✗	0.3821	0.7133	1.3427	0.8584	0.7545	0.6228	0.5742
w/ semantic	✓	✓	✓	✗	0.3945	0.7512	1.3742	0.8701	0.7742	0.6921	0.6321
	✓	✓	✗	✓	**0.4058**	**0.7776**	**1.3851**	**0.8807**	**0.7968**	**0.7147**	**0.6451**

Sample	**1**	**2**	**3**
Img Pair			
GT	a winding road is built and many houses are constructed beside it to replace the former vegetation.	a road appears at the bottom and many houses are scattered replacing the trees.	two buildings disappear.
w/o semantic	some houses have been built at the end of the road and many trees are removed.	the vegetation has been replaced by a residential district with roads and houses.	two houses appear on the bareland.
w/ semantic	a road in the shape of an arc has been constructed across the forest and many villas are scattered.	trees are replaced by a road with villas built along.	two buildings disappear in the lower left corner.

Sample	**4**	**5**	**6**
Img Pair			
GT	more houses along the new branch of the road appear on the bare ground.	two intersecting roads and some houses are built to replace a row of trees.	the row of houses above disappears and five houses appear in the middle.
w/o semantic	more houses appear on both sides of the road.	several roads with houses around appear in the bareland.	some houses appear in the center.
w/ semantic	a residential area is built along the new road at the bottom.	two roads and some houses appear at the top left corner of the scene.	the houses above are removed and some buildings appear.

Fig. 8. Impact of pixel-level CD semantic guidance on change caption generation. Red indicates incorrect entity descriptions, while green indicates correct entity descriptions.

3) The application of the cross-task attention mechanism, based on bilinear similarity, is superior to the direct use of the cross-attention mechanism in the Transformer, especially on ROUGE_L, BLEU-2, BLEU-3, and BLEU-4, which indicates that the model's improvement is centered on a comprehensive perception of changes.

To further scrutinize the advantages of pixel-level CD semantic guidance for change caption generation, we present some description generation results in Fig. 8, and the spatial response heatmaps of the differential feature map in Fig. 9. Fig. 8 reveals that by incorporating a semantic guidance branch, i.e., the MTL, the model can accurately delineate the location and quantity of changes. For instance, in subfigures 1–3, the model without semantic guidance inaccurately describes the location of changes, the number of roads, and the presence or absence of changes, whereas the model with semantic guidance can correctly perceive these semantics. Moreover, it can also better comprehend the scene, for example, in subfigure 4, the model with semantic guidance can identify the appearance of new roads; in subfigure 5, it can perceive the change in quantity and provide the location information of the change; in subfigure 6, it can describe the increase or decrease of different building instances simultaneously. Please be mindful that in Table II and Figs. 8 and 9, the configuration without semantic guidance (w/o semantic) is analogous to the single-task learning setting, whereas the setup with semantic guidance (w/ semantic) corresponds to the MTL setting.

Given that the features within the CC decoder have been deeply integrated with the text instructions, which are high-level semantic features, we visualized the bitemporal differential features postneck with heatmaps. The image pairs in Fig. 9 exhibit dramatic semantic-irrelevant changes such as various lighting and contrast conditions. From Fig. 9, it is evident that: 1) for scenes without changes, i.e., the left

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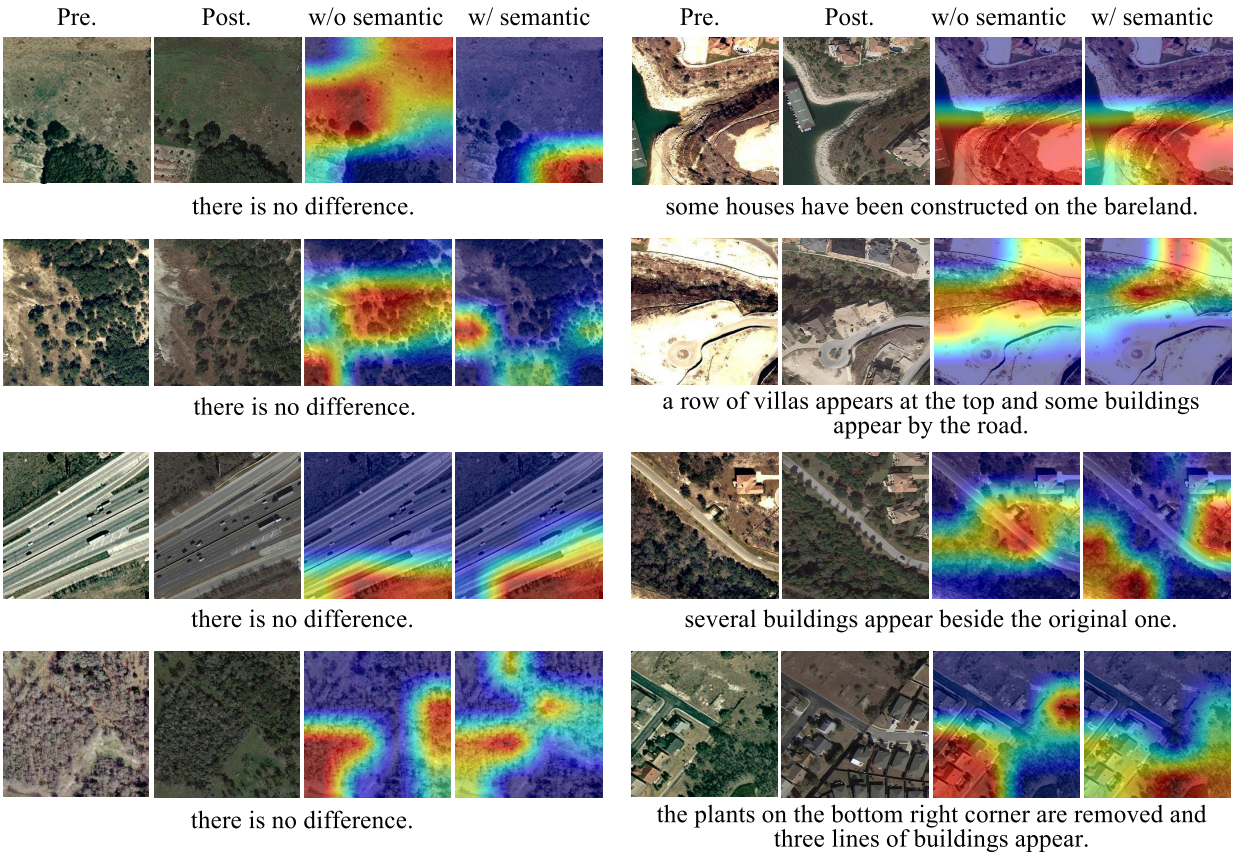


Fig. 9. Heatmap of the bitemporal differential features of pixel-level CD semantic guidance, where darker colors indicate higher feature response values.

TABLE III

PERFORMANCE OF DIFFERENT POSITIONS AND STRUCTURES OF THE CHANGE SEMANTIC FEATURE ENHANCER IN CAPTIONING DECODER ON LEVIR-CC TEST SET. PRE./POST.: THE ENHANCER BEFORE/AFTER THE Q-FORMER. ACT.: TEMPORAL FEATURE INTERACTION. SUB.: BITEMPORAL FEATURE SUBTRACTION

Pre.	Post.	Act.	Sub.	METEOR	ROUGE$_L$	CIDEr	BLEU-1	BLEU-2	BLEU-3	BLEU-4
✓	✗	✗	✓	0.3342	0.5914	1.2143	0.7941	0.6991	0.5545	0.4551
✗	✓	✗	✓	0.3483	0.6042	1.2594	0.8264	0.7072	0.6062	0.5275
✓	✗	✓	✗	0.3546	0.6547	1.2794	0.8184	0.7181	0.6174	0.5432
✗	✓	✓	✗	0.3714	0.7043	1.3053	0.8438	0.7274	0.6221	0.5621
✓	✗	✓	✓	0.3696	0.6954	1.2977	0.8384	0.7143	0.6164	0.5591
✗	✓	✓	✓	**0.3821**	**0.7133**	**1.3427**	**0.8584**	**0.7545**	**0.6228**	**0.5742**

subfigures, the model with semantic guidance can generate lower and smaller differential feature responses, indicating the ignorance on semantic-irrelevant changes and 2) for scenes with changes, i.e., the right subfigures, the model with semantic guidance exhibits more focused and accurate responses, which are essentially consistent with the areas where effective changes have occurred, while the one without semantic guidance focus on irrelevant areas.

2) Effects of the Structure of Change Semantic Feature Enhancer in CC Decoder: Given the significant differences in input and attended semantics compared to single-image visual-language models, we developed a change semantic feature enhancer for change description within the decoder. This enhancer generates bitemporal differentiated features, enabling the LLM to produce change descriptions, as depicted in (9). We evaluated the effects of the enhancer's position (before or after the Q-Former) and the enhancer's structure on the performance of the change description. For the sake of simplicity, we carried out experiments on the CC task devoid of semantic guidance. We define the first two formulas for temporal feature interaction in (9) as activation operations, and the subtraction of bitemporal features as differentiation operations. These two operations are aggregated to form various structures of the enhancer.

The experimental results are presented in Table III. The table leads us to the following observations.

1) Positioning the enhancer after the Q-Former results in optimal performance, as the pretrained Q-Former, designed to model a single image, encounters difficulties when processing differentiated features.

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TABLE IV
PERFORMANCE OF DIFFERENT TRAINING STRATEGIES ON LEVIR-CC TEST SET

Strategy	METEOR	ROUGE$_L$	CIDEr	BLEU-1	BLEU-2	BLEU-3	BLEU-4
1-stage	0.3501	0.7335	1.3053	0.8051	0.7017	0.6021	0.5622
2-stage	0.3895	0.7402	1.3591	0.8691	0.7737	0.6834	0.6223
3-stage	**0.4058**	**0.7776**	**1.3851**	**0.8807**	**0.7968**	**0.7147**	**0.6451**

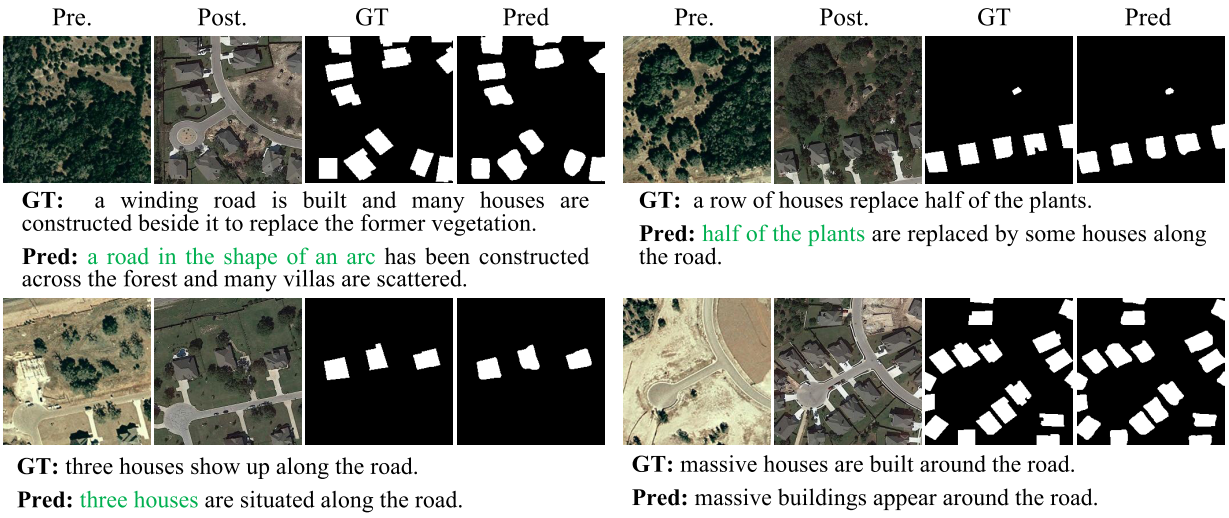


Fig. 10. Visualization results of CD and change caption on the LEVIR-CC dataset.

2) Employing solely the differentiation operation can generate differentiated features, but it fails to model the effective changes of bitemporal features and to filter out irrelevant information. Utilizing only the activation operation can model the shared semantics of bitemporal features, but it cannot directly capture differentiated features. Both of these structures lead to a certain degree of performance degradation. However, their integration within the enhancer can ignore interference and grasp changes simultaneously, to expedite network convergence and yield superior performance.

3) Effects of Different Training Strategies: To ensure a stable convergence in MTL and enable pixel-level semantic CD to guide the enhancement of CC, we conducted experiments with various training strategies. These strategies are categorized based on the frequency of dataset switches per epoch into one-stage, two-stage, and three-stage. The three-stage strategy, as detailed in Section III-F, was ultimately adopted in this study. The two-stage strategy involves initial training of CD, followed by CC (only including steps 1) and 2) described in Section III-F), while the 1-stage strategy entails a combined training of CD and CC. (only including step 3) described in Section III-F). The experimental results, as presented in Table IV, indicate that the three-stage training yields the most optimal performance in the CC task. This can be attributed to the model's initial optimization in accordance with individual objectives during the training process, followed by a joint optimization. This strategy not only alleviates tasks' coupling but also circumvents the risk into a local optimum. Furthermore, it facilitates a faster convergence speed for the target task.

4) Comparison With the SOTA on CD: In this section, we compare the output of the CC branch with other SOTA CD methods, including ChangeFormer [85], IDET [84], WNet [86], CSTSUNet [87], and TTP [3]. Notably, TTP is a CD algorithm developed based on the SAM foundation model, currently demonstrating the most superior performance on the LEVIR-CD dataset. The comparative results, as presented in Table V, reveal that Semantic-CC also exhibits outstanding performance in the CD task, achieving metrics comparable to TTP. Importantly, it does not induce negative transfer due to MTL, i.e., the learning of one task does not compromise the performance of another. Additionally, we provide several visual segmentation examples in Fig. 10, illustrating that Semantic-CC is capable of generating high-precision CD masks and CC descriptions. It is noteworthy that in the dataset utilized in this study, CD only provides segmentation masks for building changes, and the semantic vocabulary of CC is more extensive than that of CD. However, setting RSICD as an auxiliary task with pixel-level semantic guidance remains effective. RSICD compels the model to focus on areas with effective changes. In urban areas, changes in building structures are frequently accompanied by the increase or decrease of unknown categories at the pixel-level semantic represented by forests and roads. Despite utilizing only house alterations as constraints, the model is adept at directing attention to areas manifesting changes including forests and roads. Our strategy of leveraging pixel-level semantics from CD to guide

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TABLE V
COMPARISON WITH OTHER SOTA CD METHODS ON LEVIR-CD TEST SET

Method	P	R	F1	IoU	OA
IDET [84]	91.3	86.6	88.9	80.2	98.1
ChangeFormer [85]	92.1	88.8	90.4	82.5	99.0
WNet [86]	91.2	90.2	90.7	83.0	99.1
CSTSUNet [87]	92.0	89.4	90.7	83.0	99.1
TTP [3]	93.0	91.7	92.1	85.6	99.2
Semantic-CC (Ours)	**93.0**	**91.8**	**92.4**	**85.8**	**99.2**

the training of the CC task is rational, hence, the disparity in semantic vocabulary does not adversely affect their respective performances.

F. Discussion on Limitations and Future Work

The experimental results have shown that Semantic-CC obtains the ability to generate granular and precise descriptions of the changes between bitemporal remote sensing images. However, there still remain some limitations to be discussed and solved, shown as follows.

1) How RSICC benefits RSICD is still vague. Table V shows Semantic-CC makes little improvement on the RSICD task, which denotes that the contribution of RSICC to RSICD is still insufficient. Researches on whether and how RSICC enhances the performance on the RSICD task are needed.
2) RSICC and RSICD are not equivalent in the setting. To be specific, RSICD is set to binary CD for a single category. The limitations in RSICD tasks hinder the possibility of Semantic-CC further developing into a universal change interpretation model. It is worth exploring how to utilize the diverse semantic information included in change captions to perform multiclass fine-grained semantic CD based on binary CD.
3) Although the utilization of pretrained foundation models reduces the amount of data required for model training, data with both semantic and pixel labels is still insufficient. Therefore, the generalization performance of Semantic-CC needs further discussion in more practical scenarios. Moreover, although the experimental results have shown that using only one category of pixel-level semantic reinforcement for CC generation is effective, it is worth discussing whether using pixel-level semantic guidance with richer categories can make the model performance more powerful.

Although Semantic-CC has demonstrated excellent performance in both RSICC and RSICD tasks, the development of the model has not yet departed from the framework of traditional neural networks, and the understanding and processing of RSICC and RSICD tasks are primitive. At present, remote sensing images are developing toward higher spatial and temporal resolutions, providing more refined information clues for understanding remote sensing scenes from a changing perspective. In the future, the model proposed in this article will develop toward the goal of comprehensively understanding and recognizing bitemporal remote sensing images at the pixel, target, and semantic levels, ultimately constructing a universal bitemporal remote sensing image understanding architecture.

V. CONCLUSION

This article introduces Semantic-CC, a novel method for remote-sensing image CC, based on foundational knowledge and semantic guidance. By leveraging the latent knowledge inherent in foundation models, Semantic-CC enhances the model's generalization capabilities across diverse spatio-temporal scenarios. Additionally, it offers pixel-level semantic guidance by incorporating a CD task, thereby yielding more granular and precise descriptions. Semantic-CC is composed of four key components: a bitemporal SAM-based encoder for the extraction of dual-image features where a change semantic filter for bitemporal interaction; a multitask semantic aggregation neck that utilizes intra- and intertask information attention for dual-task interaction; a CD decoder that provides pixel-level semantic guidance for CC; and a Vicuna-based change caption decoder for the generation of change description sentences. To ensure the smooth convergence of CD and CC, and to prevent negative transfer in MTL, we have also devised a three-stage training strategy. Experimental results on the LEVIR-CD and LEVIR-CC datasets substantiate the complementary nature of the semantics of CC and CD. They also illustrate that Semantic-CC is capable of producing high-precision, granular CD masks and CC descriptions.

REFERENCES

[1] P. Du et al., "Remote sensing image interpretation for urban environment analysis: Methods, system and examples," *Remote Sens.*, vol. 6, no. 10, pp. 9458–9474, Oct. 2014.
[2] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.
[3] K. Chen et al., "Time travelling pixels: Bitemporal features integration with foundation model for remote sensing image change detection," 2023, *arXiv:2312.16202*.
[4] K. Chen, W. Li, J. Chen, Z. Zou, and Z. Shi, "Resolution-agnostic remote sensing scene classification with implicit neural representations," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[5] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.
[6] K. Chen et al., "Continuous remote sensing image super-resolution based on context interaction in implicit function space," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702216.
[7] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713.
[8] S. Mei et al., "Lightweight multiresolution feature fusion network for spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501414.
[9] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.
[10] S. Mei, R. Jiang, X. Li, and Q. Du, "Spatial and spectral joint super-resolution using convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4590–4603, Jul. 2020.
[11] C. Liu, R. Zhao, H. Chen, Z. Zou, and Z. Shi, "Remote sensing image change captioning with dual-branch transformers: A new method and a large scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633520.
[12] G. Hoxha, S. Chouaf, F. Melgani, and Y. Smara, "Change captioning: A new paradigm for multitemporal remote sensing image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627414.

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[13] C. Liu, K. Chen, B. Chen, H. Zhang, Z. Zou, and Z. Shi, "RSCaMa: Remote sensing image change captioning with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[14] C. Liu, R. Zhao, J. Chen, Z. Qi, Z. Zou, and Z. Shi, "A decoupling paradigm with prompt learning for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. –18, 2023, Art. no. 5622018.

[15] S. Chang and P. Ghamisi, "Changes to captions: An attentive network for remote sensing change captioning," *IEEE Trans. Image Process.*, vol. 32, pp. 6047–6060, 2023.

[16] X. Shen, B. Liu, Y. Zhou, J. Zhao, and M. Liu, "Remote sensing image captioning via variational autoencoder and reinforcement learning," *Knowl.-Based Syst.*, vol. 203, Sep. 2020, Art. no. 105920.

[17] C. Liu, K. Chen, Z. Qi, H. Zhang, Z. Zou, and Z. Shi, "Pixel-level change detection pseudo-label learning for remote sensing change captioning," 2023, *arXiv:2312.15311*.

[18] M. Wang et al., "RSBuilding: Towards general remote sensing image building extraction and change detection with foundation model," 2024, *arXiv:2403.07564*.

[19] C. Liu, K. Chen, H. Zhang, Z. Qi, Z. Zou, and Z. Shi, "Change-agent: Toward interactive comprehensive remote sensing change interpretation and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635616.

[20] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spatial Inf. Sci.*, vol. 26, no. 3, pp. 262–288, Jul. 2023.

[21] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[22] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 35, 2022, pp. 23716–23736.

[23] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.

[24] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–11.

[25] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.

[26] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[27] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[28] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[29] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[30] Z. Chen et al., "Intern VL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 24185–24198.

[31] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[32] D. Wang et al., "MTP: Advancing remote sensing foundation model via multitask pretraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11632–11654, 2024.

[33] X. Lu, J. Yuan, R. Niu, Y. Hu, and F. Wang, "Viewpoint integration and registration with vision language foundation model for image change understanding," 2023, *arXiv:2309.08585*.

[34] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Remote sensing image change detection with Mamba," 2024, *arXiv:2406.04207*.

[35] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[36] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711.

[37] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[38] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[39] F. Rahman, B. Vasu, J. V. Cor, J. Kerekes, and A. Savakis, "Siamese network with multi-level features for patch-based change detection in satellite imagery," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 958–962.

[40] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

[41] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[42] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[43] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[44] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[45] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2020.

[46] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[47] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[48] C. Ren, X. Wang, J. Gao, X. Zhou, and H. Chen, "Unsupervised change detection in satellite images with generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10047–10061, Dec. 2021.

[49] C. Liu, R. Zhao, and Z. Shi, "Remote-sensing image captioning based on multilayer aggregated transformer," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[50] S. Chouaf, G. Hoxha, Y. Smara, and F. Melgani, "Captioning changes in bi-temporal remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2891–2894.

[51] C. Liu, J. Yang, Z. Qi, Z. Zou, and Z. Shi, "Progressive scale-aware network for remote sensing image change captioning," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 6668–6671.

[52] Y. Zhang and Q. Yang, "A survey on multi-task learning," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 12, pp. 5586–5609, Mar. 2021.

[53] S. Vandenhende, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Trans. pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3614–3633, Jan. 2021.

[54] P. Liu, X. Qiu, and X. Huang, "Recurrent neural network for text classification with multi-task learning," 2016, *arXiv:1605.05101*.

[55] M.-T. Luong, Q. V. Le, I. Sutskever, O. Vinyals, and L. Kaiser, "Multi-task sequence to sequence learning," 2015, *arXiv:1511.06114*.

[56] S. Liu, E. Johns, and A. J. Davison, "End-to-end multi-task learning with attention," in *Proc. CVPR*, Jun. 2019, pp. 1871–1880.

[57] T. Standley, A. Zamir, D. Chen, L. Guibas, J. Malik, and S. Savarese, "Which tasks should be learned together in multi-task learning?" in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9120–9132.

[58] A. R. Zamir et al., "Robust learning through cross-task consistency," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11194–11203.

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[59] K. Chen et al., "OvarNet: Towards open-vocabulary object attribute recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23518–23527.

[60] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12299–12310.

[61] E. Mohamed and A. E. Sallab, "Spatio-temporal multi-task learning transformer for joint moving object detection and segmentation," in *Proc. IEEE Int. Intell. Transp. Syst. Conf. (ITSC)*, Sep. 2021, pp. 1470–1475.

[62] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.

[63] M. Hosseinzadeh and Y. Wang, "Image change captioning by learning from an auxiliary task," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2725–2734.

[64] C. Xu, R. T. Tan, Y. Tan, S. Chen, X. Wang, and Y. Wang, "Auxiliary tasks benefit 3D skeleton-based human motion prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 9475–9486.

[65] K. E. Ak, Y. Sun, and J. H. Lim, "Learning by imagination: A joint framework for text-based image manipulation and change captioning," *IEEE Trans. Multimedia*, vol. 25, pp. 3006–3016, 2022.

[66] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "Geochat: Grounded large vision-language model for remote sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2024, pp. 27831–27840.

[67] Y. Hu, J. Yuan, C. Wen, X. Lu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," 2023, *arXiv:2307.15266*.

[68] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[69] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[70] F. Yao et al., "RingMo-sense: Remote sensing foundation model for spatiotemporal prediction via spatiotemporal evolution disentangling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5620821.

[71] D. Wang et al., "Advancing plain vision transformer toward remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5607315.

[72] X. Li, C. Wen, Y. Hu, and N. Zhou, "RS-CLIP: Zero shot remote sensing scene classification via contrastive vision-language supervision," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103497.

[73] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," 2023, *arXiv:2306.11300*.

[74] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26286–26296.

[75] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[76] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.

[77] Y. Li, H. Mao, R. Girshick, and K. He, "Exploring plain vision transformer backbones for object detection," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Oct. 2022, pp. 280–296.

[78] L. Zheng et al., "Judging llm-as-a-judge with mt-bench and chatbot arena," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–9.

[79] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics*, 2002, pp. 311–318.

[80] S. Banerjee and A. Lavie, "Meteor: An automatic metric for mt evaluation with high levels of correlation with human judgments," in *Proc. ACL-WMT*, 2004, pp. 65–72.

[81] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Proc. Text Summarization Branches Out*, 2004, pp. 74–81.

[82] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2015, pp. 4566–4575.

[83] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

[84] Q. Guo, R. Wang, R. Huang, S. Sun, and Y. Zhang, "IDET: Iterative difference-enhanced transformers for high-quality change detection," 2022, *arXiv:2207.09240*.

[85] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IGARSS*, Jul. 2022, pp. 207–210.

[86] X. Tang, T. Zhang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "WNet: W-Shaped hierarchical network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615814.

[87] Y. Wu et al., "CSTSUNet: A cross Swin transformer-based Siamese U-shape network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623715.



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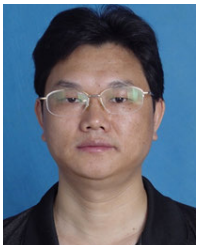
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